PROJECT ALLOTMENT LIST

ERP	Name	Sem-Sec	PS No
RU-26-02040	Anant Belchandan	1st Sem 1A	41
RU-26-01937	DEVANSHI PAPIHE	1st Sem 1A	42
RU-26-01926	kajal Kumari	1st Sem 1A	43
RU-26-01920	Vanshika Vishwakarma	1st Sem 1A	44
RU-26-01795	Diksha Dhuria	1st Sem 1A	45
RU-26-01783	Devanshu Kumar Tandan	1st Sem 1A	46
RU-26-01778	Aaditya Raj	1st Sem 1A	47
RU-26-01744	HAMISHA SAHU	1st Sem 1A	48
RU-26-01745	Deeksha Sharma	1st Sem 1A	49
RU-26-01718	damini sori	1st Sem 1A	50
RU-26-01999	SUDHANSHU KUMAR	1st Sem 1A	51
RU-26-01895	Anshika Singh Rajput	1st Sem 1A	52
RU-26-01754	Ansh Kumar Chaudhary	1st Sem 1A	53
RU-26-01590	Abhishek Kumar Bharti	1st Sem 1A	54
RU-26-01554	AKASH TIRKEY	1st Sem 1A	55
RU-26-01506	Arya Upadhyay	1st Sem 1A	56
RU-26-01445	Manav kumar singh	1st Sem 1A	57
RU-26-01426	ayush kumar sahu	1st Sem 1A	58
RU-26-00764	Abhishek Kumar Ekka	1st Sem 1A	59
RU-26-00557	Madhav Kumar	1st Sem 1A	60

PS No.	42
Project Title	**Personal Expense & Budget Manager**
Problem Statement	Build a tool that stores expenses in a CSV file, lets the user set a monthly budget, and reports category-wise spending, the remaining budget and warnings when the budget is almost used or exceeded. **Language:** Python 3 **Concepts:** Functions, Dictionaries, CSV, Pandas **Note:** Expenses are saved in expenses.csv with columns Date, Category, Amount. Menu: 1 Set budget, 2 Add expense, 3 Category summary, 4 Report. A warning is shown when 80% of the budget is used and 'Budget exceeded' above 100%.
Test Cases / Case Statements (4 cases)	

INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
Menu option 1 (Set budget) Budget: 5000	Monthly budget set: 5000
Test Case 2: Boundary / Minimum Input	
Budget: 1000 Menu option 2 (Add) 2026-10-05, Food, 1000	Expense saved. Total spent: 1000 | Remaining: 0 Budget fully used (100.00%).
Test Case 3: Invalid Input	
(a) Add: 2026-10-05, Food, -200 (b) Add: 05-10-2026, Food, 200	(a) Invalid amount. Amount must be a positive number. Expense not saved. (b) Invalid date. Use YYYY-MM-DD.
Test Case 4: Different Valid Scenario	
Budget: 5000 Expenses: Food 1500, Travel 1000, Food 1700 Menu option 4 (Report)	Food: 3200 | Travel: 1000 Total spent: 4200 | Remaining: 800 Warning: 84.00% of the budget used.

Bare Minimum Requirements	1. Write the solution in Python 3 (one .py file) and run it from the console. 2. Store and read the expenses using a CSV file and a Pandas DataFrame. 3. Use groupby() to show the category-wise spending. 4. Calculate the total spent, the remaining budget and the percentage of the budget used. 5. Show a warning at 80% and a 'Budget exceeded' message above 100%, and validate the date and the amount.
Instruction	1. Understand the problem statement and define the problem clearly. 2. Develop the solution using proper logic and programming concepts. 3. Implement all required features and pass the given test cases. 4. Test and debug the project to ensure correct output. 5. Prepare the documentation/report and upload the code and report to GitHub. 6. Complete and submit the project within the given deadline.